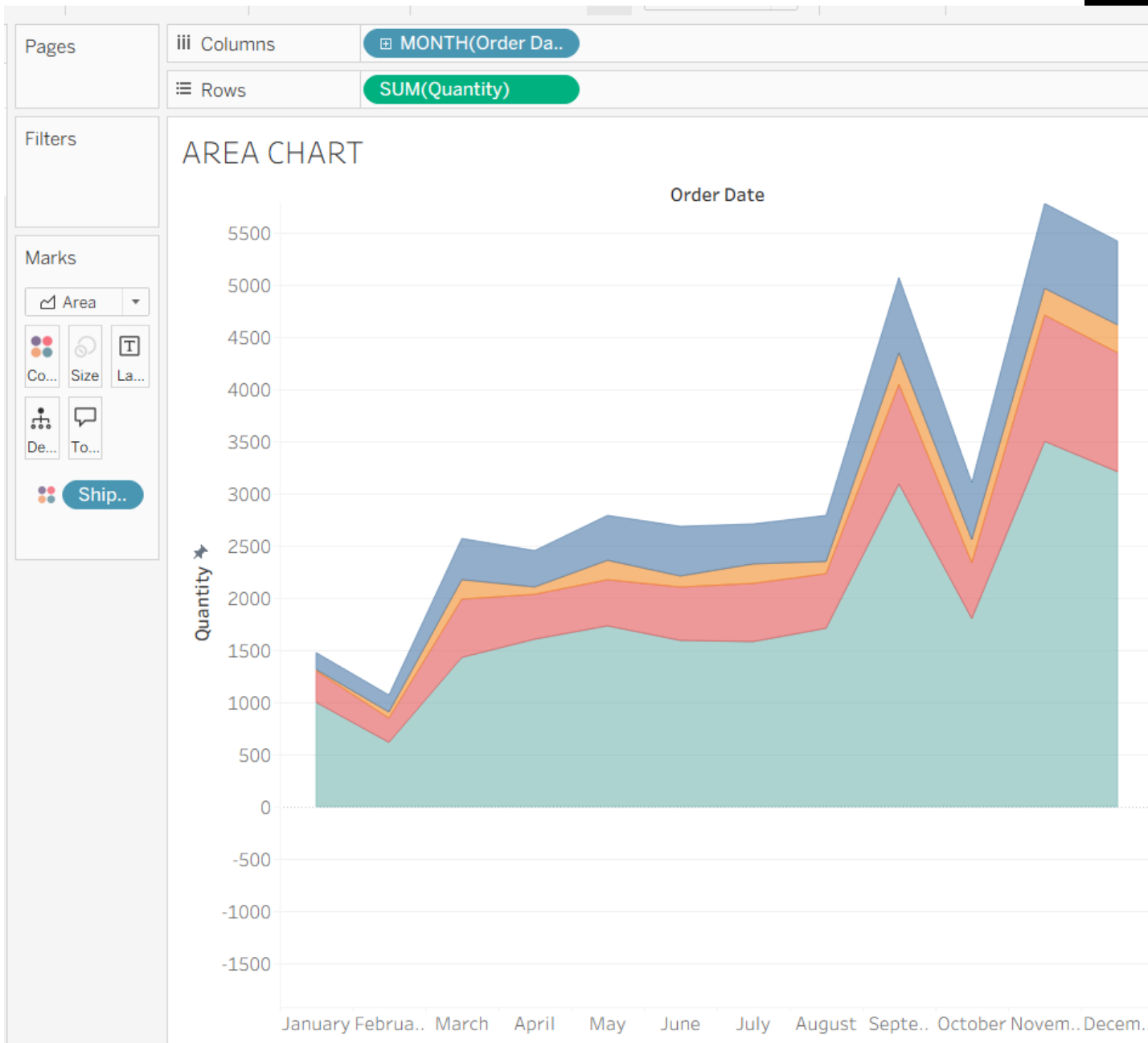


Assignment

Sheet – 01 – Area Chart

- * There will be **date pane** in left side , drag **Order date** and put it in **the columns shelf**
- * After right click on the YEAR (**ORDER DATE**) on **the columns shelf**.
- * Then select month
- * **In date pane**, drag **QUANTITY** to the **rows shelf**
- * Drag the **ship mode** to the mark card and add color on the **mark card**
- * On the **mark card** ,select the drop -down menu and select the area **the chart type**



Sheet – 02 – Bar Chart

- 1. To **rows shelf**, drag **sales** and place it.
 - * “ then the sales will be **SUM (Sales)** in the **tableau** ,which will be automatically aggregates in tableau “
 - * The **Order Date** is used ,Tableau first shows a **LINE CHART** which is representing yearly sales trends.
- 2. Click the drop-down menu on the **mark card** ,then **select Bar**.
- 3. Add the **SHIP MODE COLORS** .

colors .

5. Drag the **Region** and place it in the **Rows shelf** on the left of **sales** .

FILTER ONLY WEST REGION

6. Again drag the **region** to the **filters shelf**

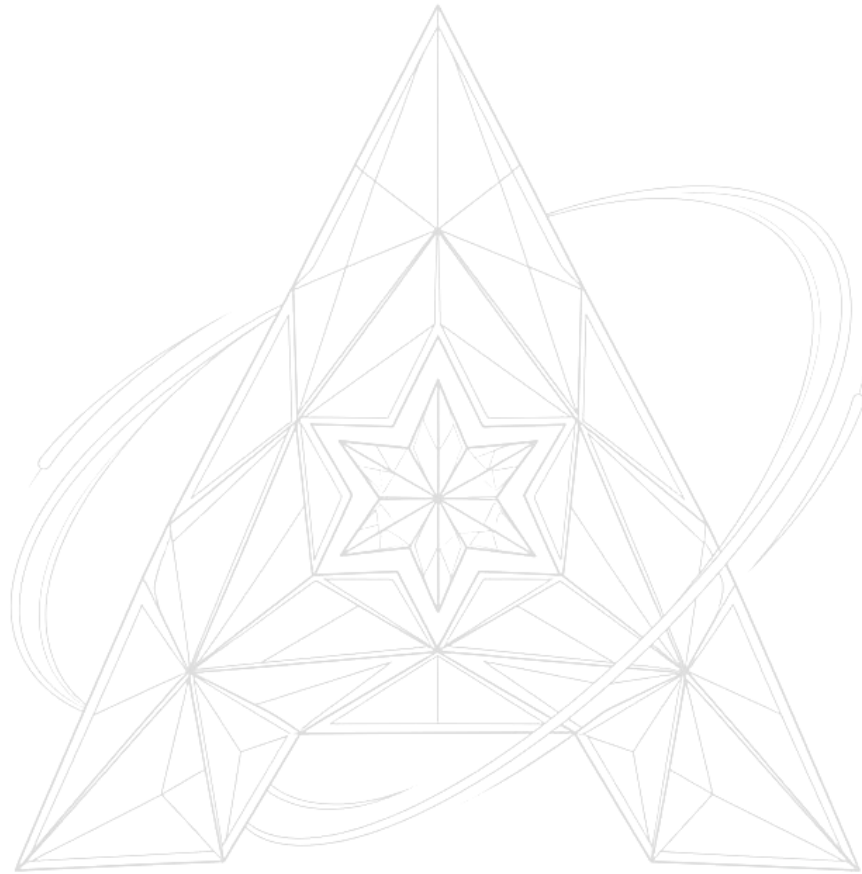
* in **filter dialog box** :

- **Uncheck Central**
- **Uncheck East**
- **Uncheck South**
- **Keep only West**

7 . click **ok** [done]

8 **Output:**

9



ALPHA
— INNOVATORS —



Pages

Columns

Rows

Region

SUM(Sales)

Filters

Region: West

Marks

Bar

Color

Size

Label

Detail

Tooltip



Ship Mode

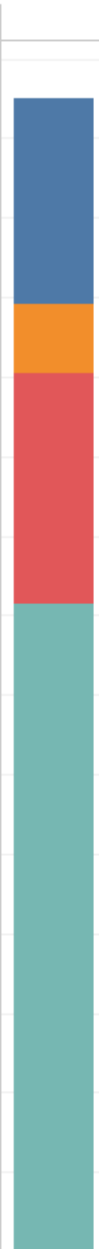
BAR CHART

Region

West

Sales

750K
700K
650K
600K
550K
500K
450K
400K
350K
300K
250K
200K
150K
100K
50K
0K



1. Drag the **segment** ,place it in the **columns shelf**

2. Drag the **Discount** ,place it in the **Rows shelf**

Change Aggregation to Average:

3. On the **Rows shelf**, click **Discount**, then hover over **Measure (Sum)** and change the aggregation to **Average**.

4. Adding **Region** , drag **region** to **columns shelf** . dropping it **right of segment**

5. **covert to box plot**

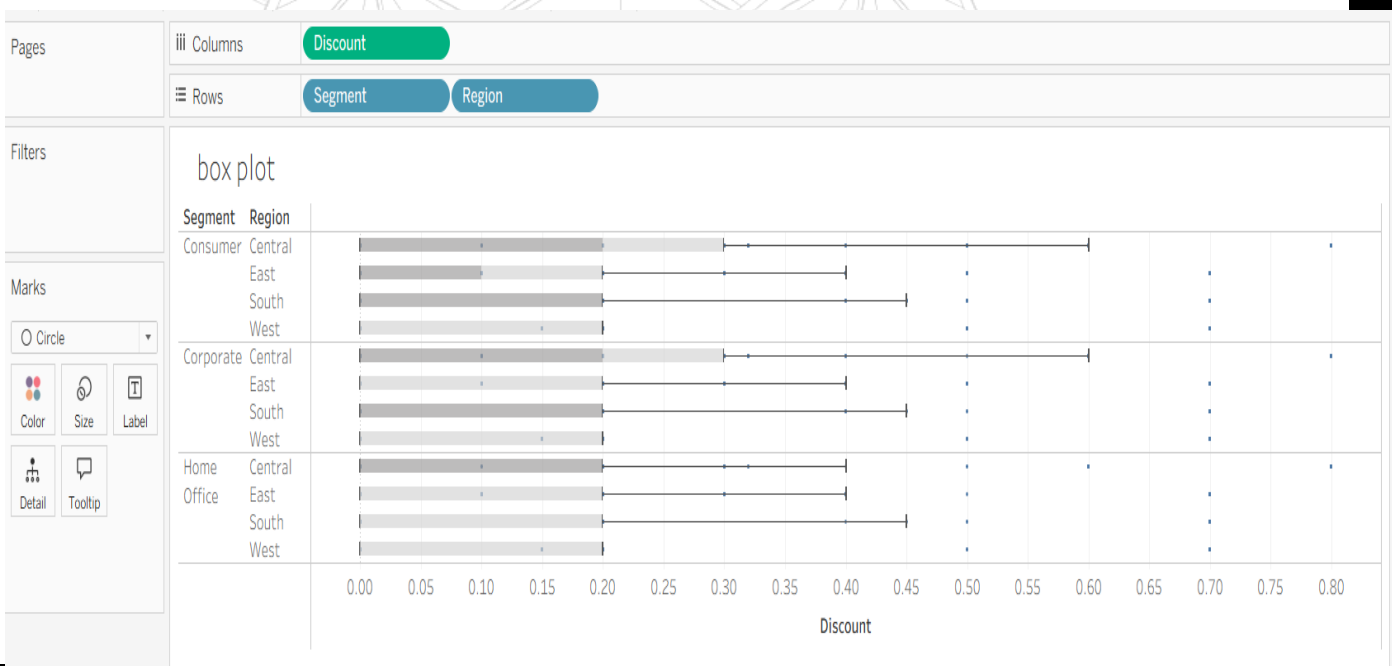
6. click **show me in toolbar** and select **box-and-whisker plot**

Region from Marks card back to columns shelf, right of segment.

8.Disaggregate Date : Now go to **analysis** and **uncheck Aggregate Measures**

9.Swap Axes(Horizontal View) : Now **click swap rows and columns button** .

Output:



Sheet – 04 - Gantt Chart

10 1. After adding **order date** to columns and tableau automatically shows it as **YEAR(Order Date)**

11 2. Now click on the year and change it to **week number**, now it will be the **weekly timeline chart**

3. Drag **sub-category** to the **rows shelf** and now drag **ship mode** to **rows shelf** then place it right side of **the sub-category**

4. now rows will shows as

* **The furniture**

– **First class** –

Standard class etc.....



bars appear across timeline.

6. Now next step **analysis to create calculated field**

Name: **OrderUntilShip**

By using formula of

DATEDIFF('day',[Order Date],[Ship Date])

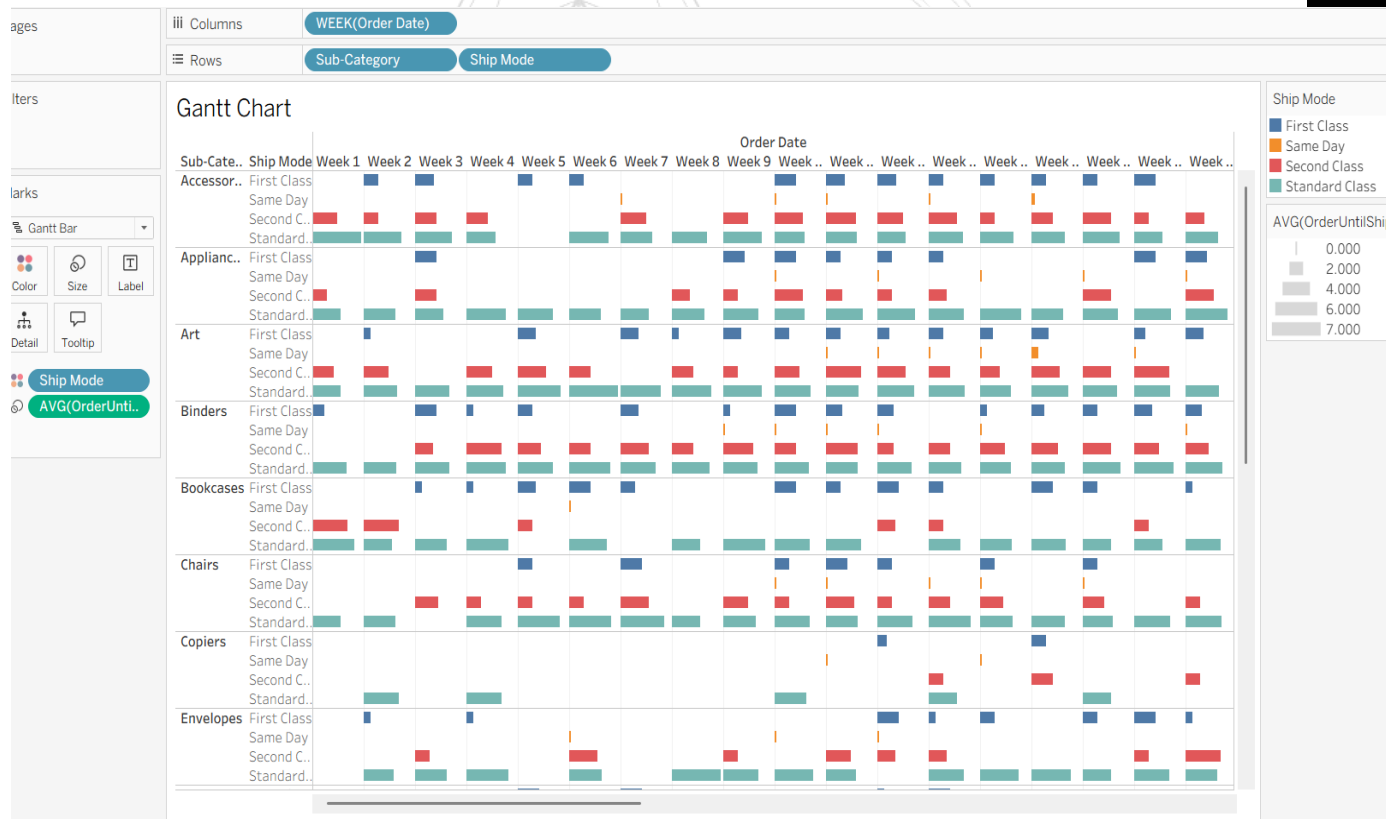
And click **okay** or **apply**

7. Next, drag **OrderUntilShip** to size on the **marks card** and Now bars will become longer based on shipping days

8. on mark card right click on **SUM(OrderUntilShip)** and select **measure** to **average**



mark card and now ship mode gets
different color
Output:



ALPHA

— INNOVATORS —



Sheet – 05 - Heat Map

1. Now by adding **segment to columns shelf** and now you will see

- * **consumer**
- * **corporate**
- * **home office**

2. drag **region to rows shelf** and **sub-category** to **row shelf** and place it beside **right of region**

3. Drag **profit** and **drop** it in the mark card on **color**

4. Now adding **border to cells** by click **color** on **mark card**, then under **Border**, choose **medium gray** to make each **box clearer**

arrow and select **Edit color**

In the dialog box

***chose palette ---->Red-black diverging**

***check Use full Color Range and click Apply or Okay**

Now the **profit** will be **black/dark**

And **loss will be red**

6.Drag **sales measure** and drop it on **the SIZE** in the mark card

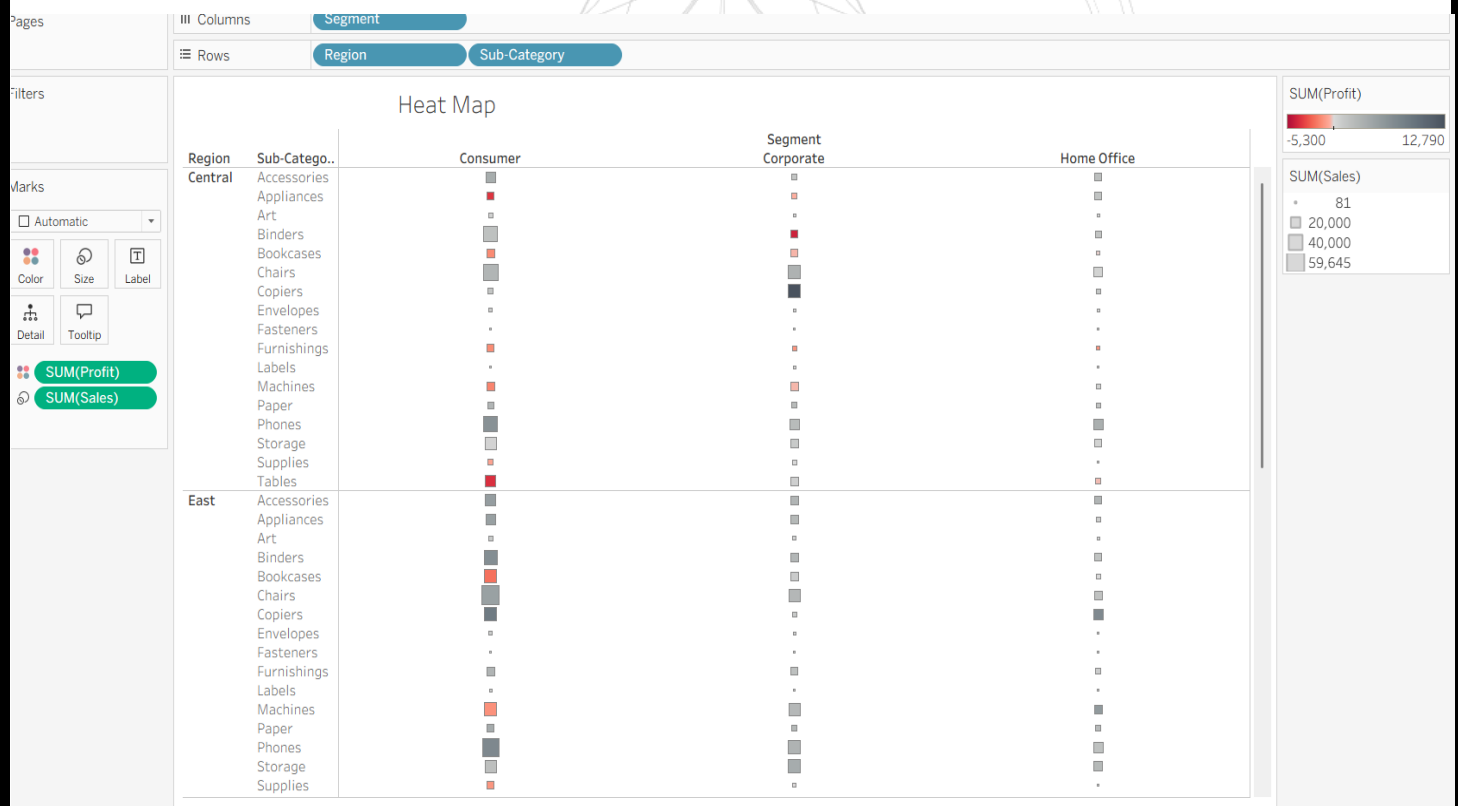
***the box size of sales amount represents**

Large box =high sales

Small box =low sales

card and move it slider to right until the boxes fit properly in the chart to get out put

Output:



ALPHA
- INNOVATORS -



1. Now drag **QUANTITY** and place it on **columns shelf**

2. on top **right corner**, click show me and select **Histogram**

Now **tableaus** will automatically does :

- * On columns creates **Quantity (bin)**
- * **Quantity** will moves to **Rows**
- * **CNT (QUANTITY)** changes aggregation

3. Now **x-axis: quantity**

Y-axis :count of orders

Ex: the quantity had orders of around
4800

— INNOVATORS —



2400

4. Drag **segment** from dimensions and drop it on color in **marks card**

Now bars becomes stacked by showns as

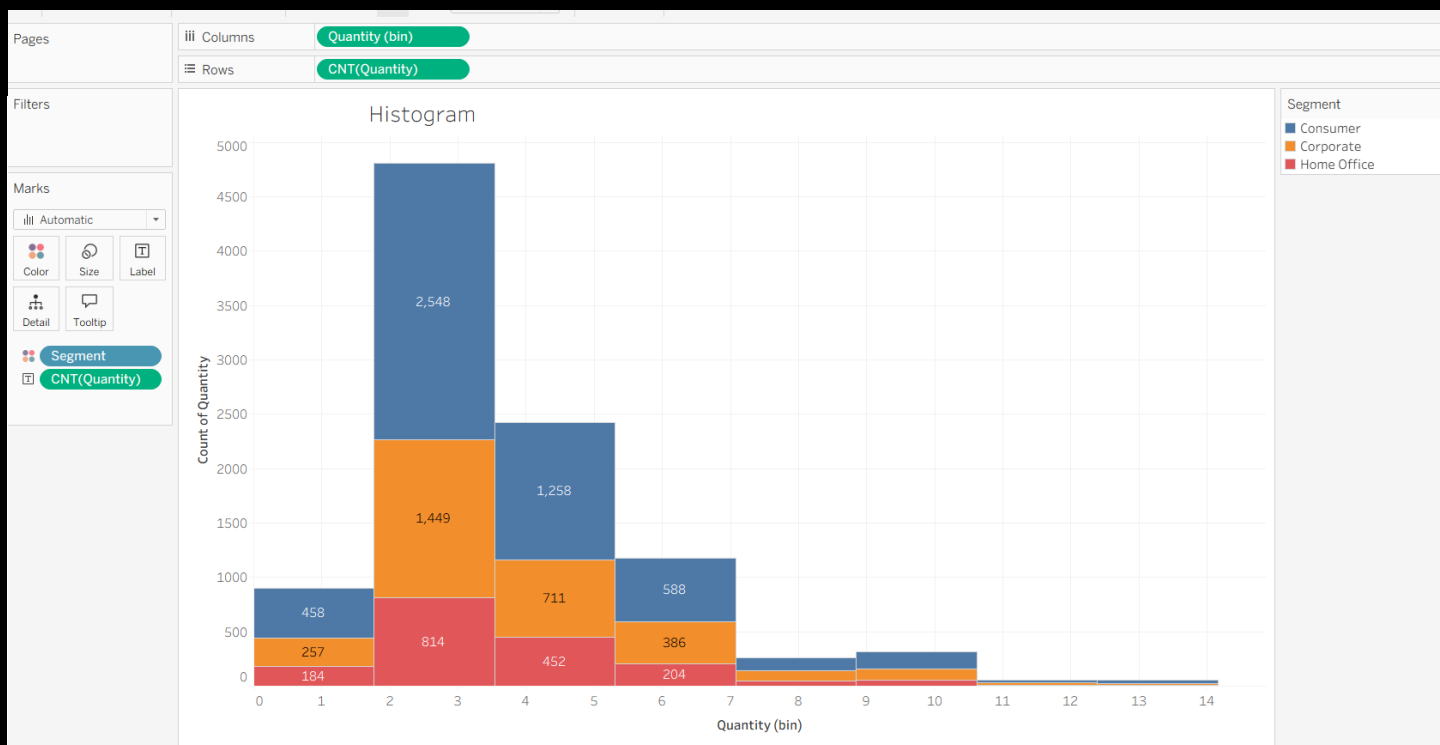
- *** consumer**
- ***corporate**
- ***Home office**

5. drag **CNT(Quantity)** from **rows shelf** to label by using **Ctrl** .

6. On mark card right click **CNT(Quantity)** then select quick table calculation to percent of total

Now labels show percentages.

Output:



ALPHA

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1. Drag **ship Date** and place it on **columns shelf**

Tableau shows: **YEAR (SHIP DATE)**

2. Now drag **sales** and place it on the **rows** shelf, then a **line chart** will appear

3. Drag **profit** and drop it in the **rows shelf**, right side of the **sales**

* **creates two separate axes:**

Sales axis

Profit axis

4. Drag **SUM(Profit)** from **Rows shelf** and place it directly on the **sales axis**.

5. Now **profit** and **sales** share **same axis**

6. Now

Repeating yearly spikes

Seasonal business growth

And this is called **seasonality**.

7. in **columns shelf** click **month** and change it to the **year** then go to **analysis pane** and add **forecast** into **chart area** and ,next change it back **year to month**.

Output:

ALPHA
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Dashboard

Layout <

Device Preview

Size

min 420x560 - max 65...

Sheets

- Sheet 2
- Sheet 3
- Sheet 4
- Sheet 5
- Sheet 6
- Sheet 7
- Sheet 8
- Sheet 9
- Sheet 10
- Sheet 11
- Sheet 12

Objects

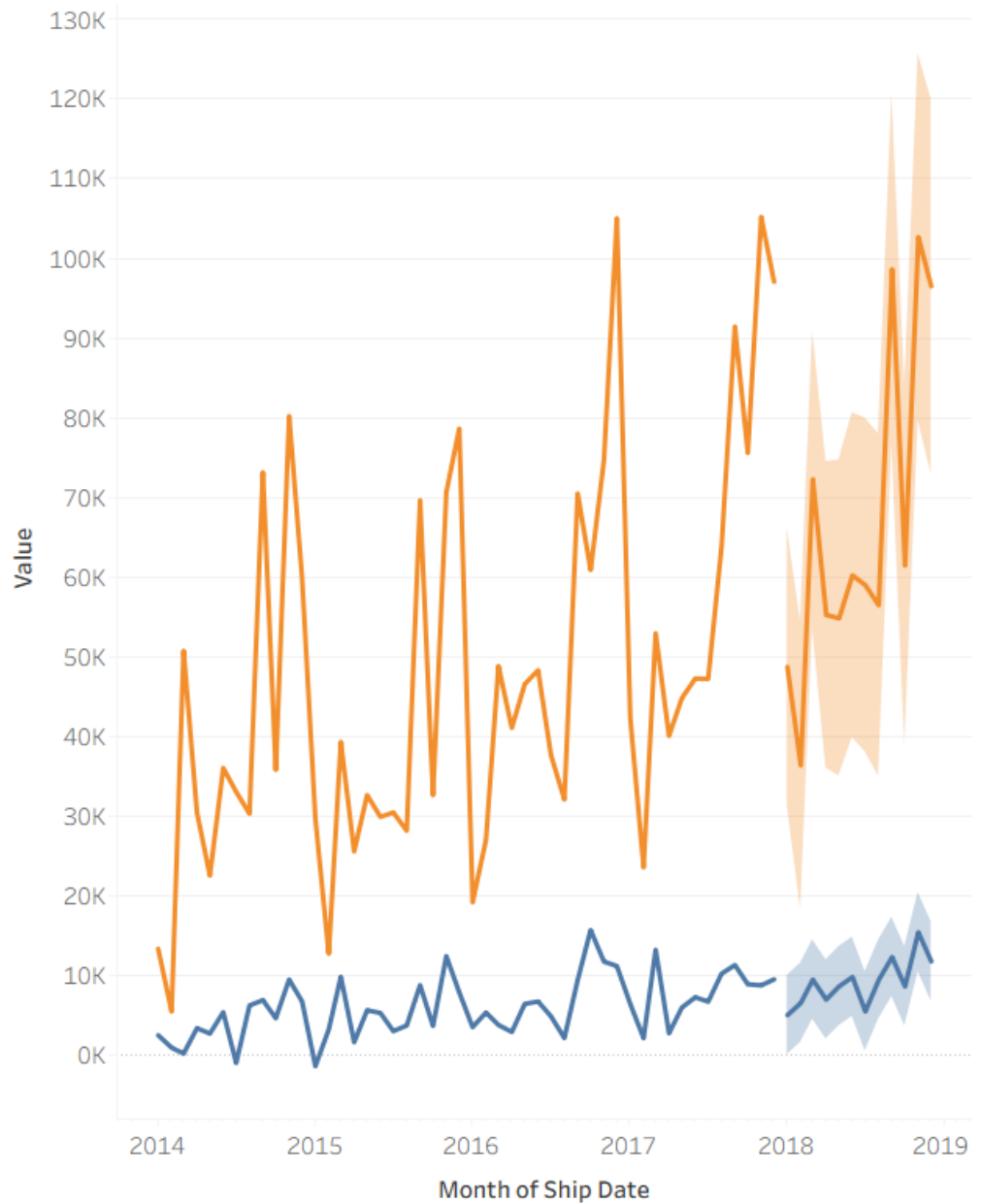
- Text
- Extension
- Pulse Metric
- Image
- Blank
- Workflow

Tiled

Floating

☒ Show dashboard title

Line chart



— INNOVATORS —

1. Adding **category** to **columns shelf**

*now, you will see

Furniture

Office supplies

Technology

2. Drag **sales** and place it on the **Rows shelf**

Bar chart create by **default**.

3. Select the show me on top right then select **packed bubbles** .then **bubbles** creates the

Furniture

Office supplies

Technology



4. Drag **region** and place it on the details in the **marks card**, now each category

Now each category splits into:

***East**

***West**

***central**

***south**

Now more bubbles will appear.

5. Drag **profit** and place it in the **color**, now bubble color represents profit

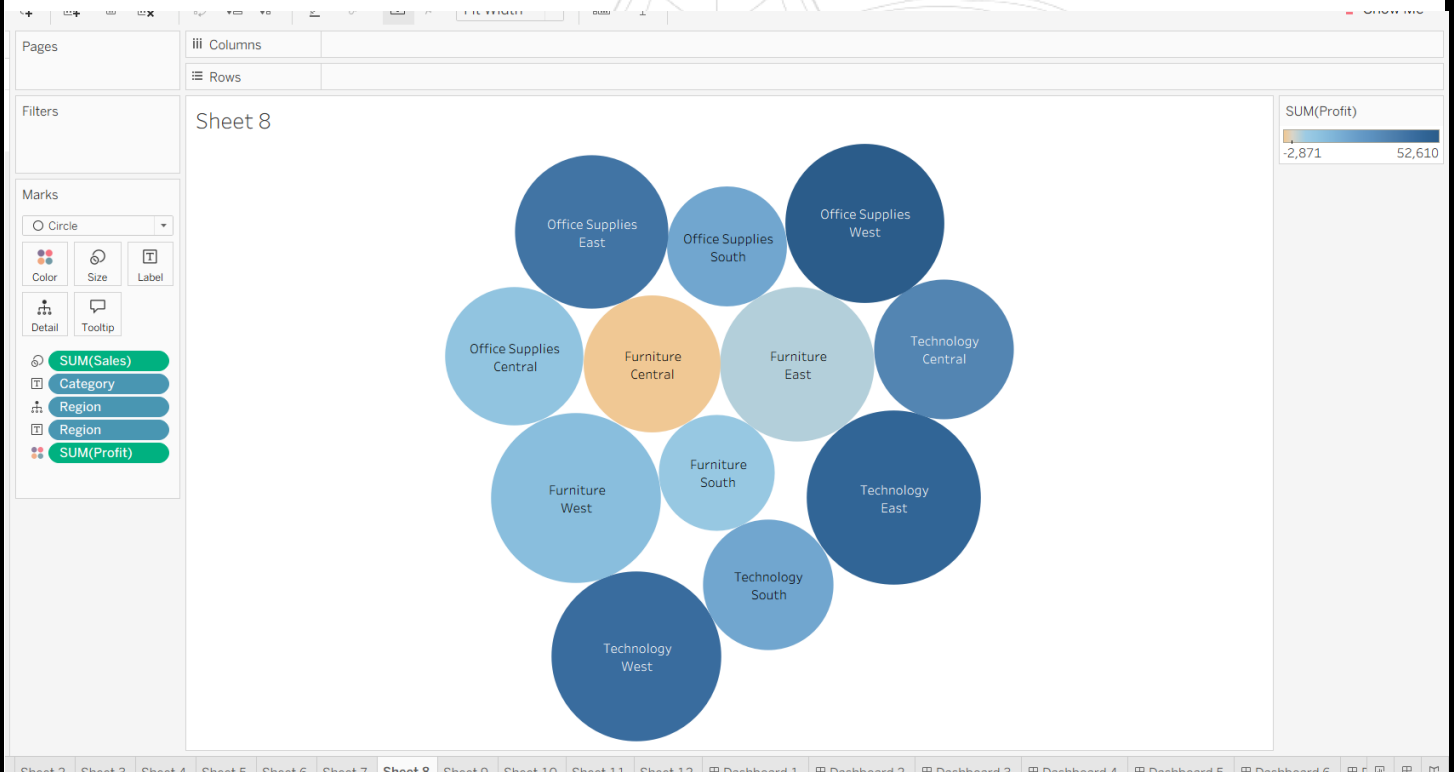
Ex : * **Dark blue =High profit**

● **Light blue =Lower profit**



Now each bubble displays region name.

Output:



ALPHA
- INNOVATORS -



1. Drag **profit** and place it on the **columns shelf**

2. Drag **category** and place it on **the Rows shelf**

Creates a **Bar chart** by default

3. click on **show me** and select the **pie chart**

4. After selecting **pie chart**

Mark the type changes **to pie**

Sum(profit) moves to:

***Angle**

***size**

Category move to

***color**

5. Now increase **pie chart** :



***select Entire view**

*** now, click size on mark card**

*** move slider right**

Now the **pie chart** becomes **larger**

6. Drag **category** and place it on the **label on mark card**.

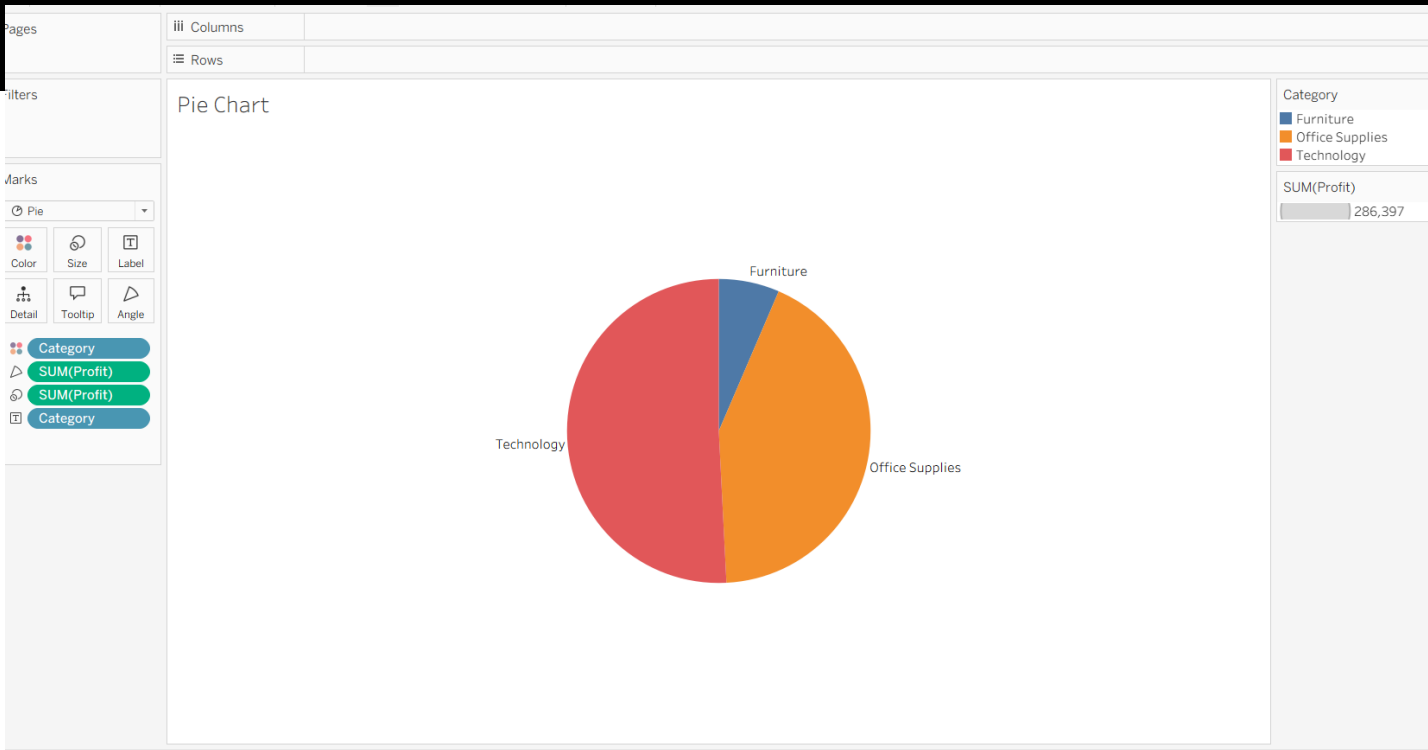
Now chart displays

- ***Furniture**
- ***Office supplies**
- *** Technology**

Output:

ALPHA
— INNOVATORS —
—





ALPHA

— INNOVATORS —



1. Drag **sales** and place it on the **columns shelf**

Now , tableau creates **horizontal axis**.

2. Drag **profit** and place it on the **Rows shelf**

Now, tableau creates **vertical axis**.

Now ,you get **one mark scatter plot** .

3. Drag **category** and place it on the **color** in **mark card** .

Now marks split into:

- **Furniture**
- **Office supplies**
- **Technology**

Each category gets different color .

card

Now more marks appear and each category is divided by

***Central**

***East**

***South**

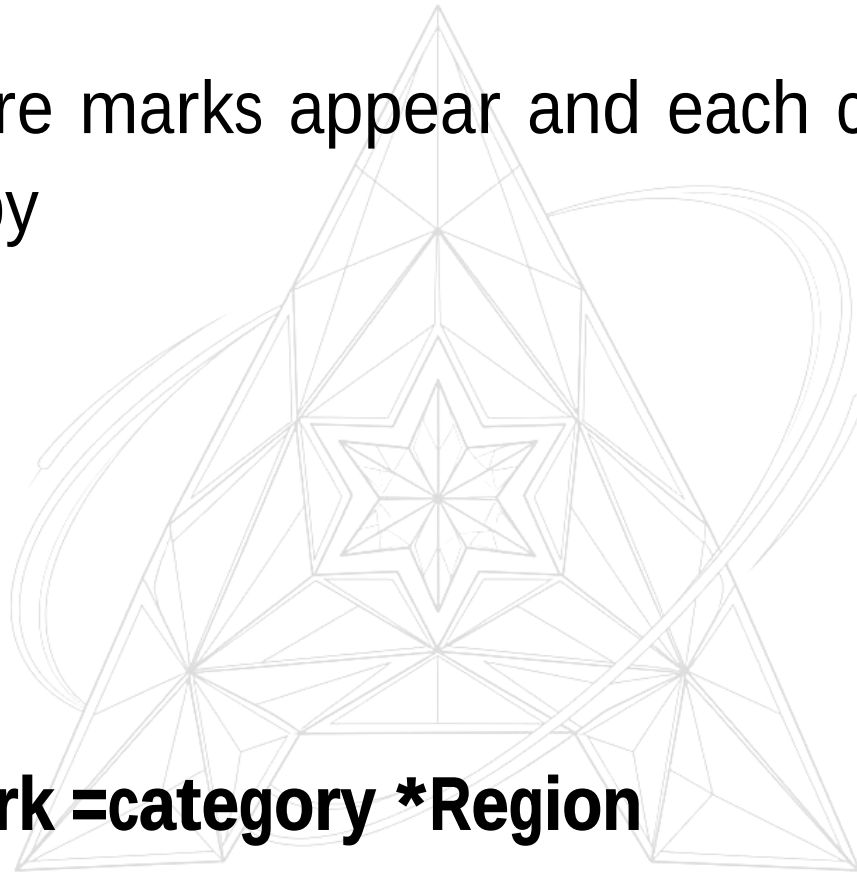
***West**

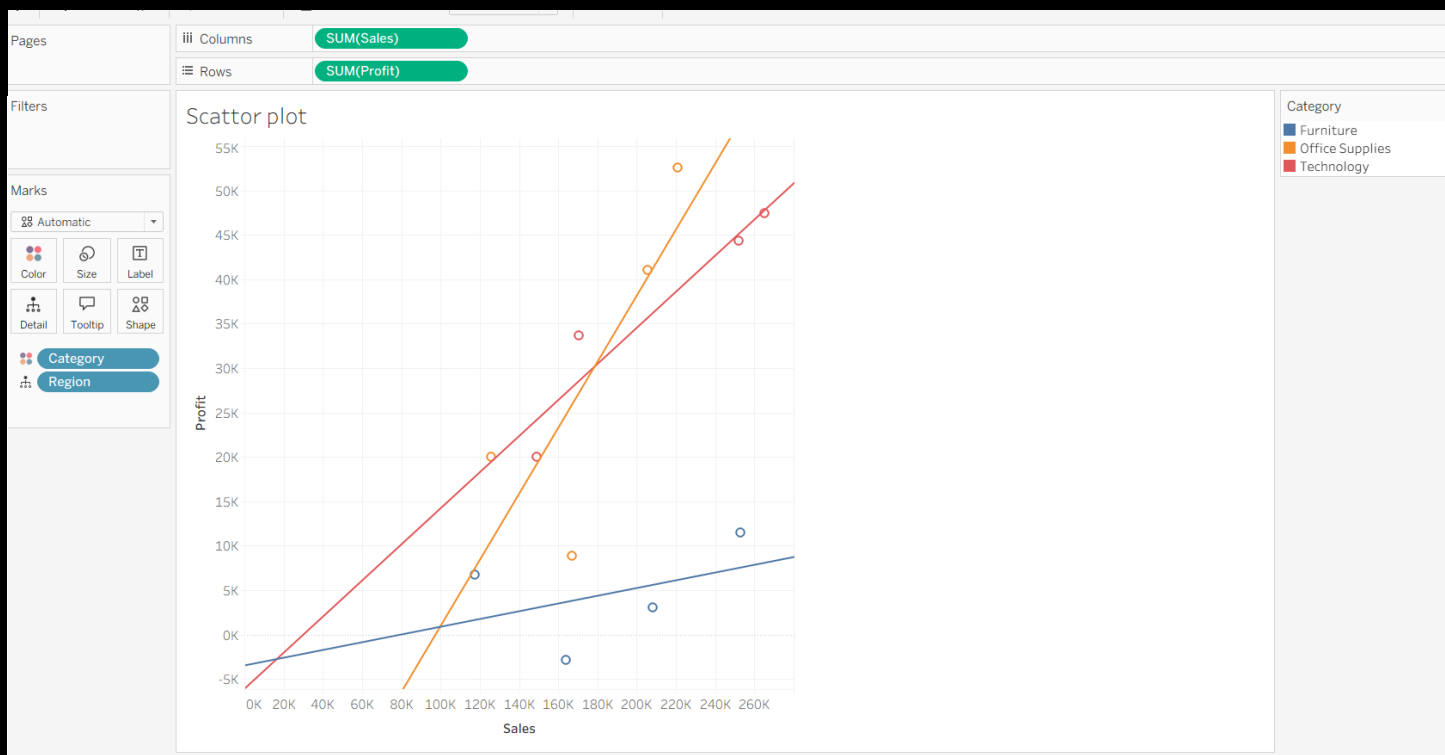
Total mark = category * Region

5. Now, click **Analytics Pane** and **Drag Trend Line** . now drop it on the chart

Then, select: **Linear**

Output:





ALPHA
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1. Drag **Order Date** and place it on the **columns shelf**

2. Drag **Sub -category** and place it on the **Rows shelf**

Now a **table structure is created.**

Rows=sub-categories

Columns=Years

3. Drag **sales** and place it on **the text in mark card**

Now, each cell shows:

SUM(sales)

For each year and sub-category .



the **sub-category in rows shelf**

Now rows become:

***Central--->Chairs**

***Central--->Phones**

***East--->Chairs**

***East--->Phones etc...**

5. Now, **right -click SUM(Sales)** on **mark card**

Select add **table calculation**

6. Configure table calculation

In dialog box

Calculation type: percent of total

Compute using : Pane (down)



Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

SUM(Sales)

Columns

YEAR(Order Date)

Rows

Region

Sub-Category

Text Table with Percentages

		Order Date			
Region	Sub-Catego..	2014	2015	2016	2017
Central	Accessories	4.27%	7.58%	7.33%	7.42%
	Appliances	3.52%	4.84%	4.08%	6.07%
	Art	0.79%	1.10%	1.03%	1.56%
	Binders	15.28%	5.73%	9.53%	14.35%
	Bookcases	1.77%	8.07%	5.69%	3.83%
	Chairs	19.99%	17.41%	15.84%	15.78%
	Copiers	3.15%	12.45%	11.87%	2.50%
	Envelopes	1.54%	0.85%	0.66%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.47%	3.45%
	Labels	1.01%	0.30%	0.35%	0.40%
	Machines	15.69%	1.80%	1.80%	4.08%
	Paper	2.26%	3.45%	3.64%	4.24%
	Phones	9.56%	18.82%	13.50%	15.78%
	Storage	10.68%	8.10%	8.69%	9.31%
	Supplies	0.42%	0.32%	2.91%	3.00%
Tables	7.50%	6.67%	9.44%	7.20%	
East	Accessories	4.70%	11.46%	3.45%	6.96%
	Appliances	4.49%	4.28%	5.22%	5.77%
	Art	1.00%	1.09%	1.04%	1.22%
	Binders	4.93%	9.09%	10.49%	6.57%
	Bookcases	8.44%	12.57%	3.30%	3.44%
	Chairs	17.10%	12.80%	12.74%	14.66%
	Copiers	3.19%	3.17%	10.94%	11.45%
	Envelopes	1.05%	0.56%	0.72%	0.40%
	Fasteners	0.08%	0.12%	0.19%	0.09%
	Furnishings	2.92%	3.37%	5.31%	4.91%
	Labels	0.40%	0.33%	0.43%	0.37%
	Machines	11.37%	9.23%	13.96%	5.55%
	Paper	3.04%	2.54%	2.73%	3.45%
	Phones	16.08%	14.43%	11.78%	16.93%
	Storage	12.59%	8.51%	12.23%	9.39%

1. Drag **Sub-category** and place it on the **columns**

Now, you will see **category names** on **horizontal axis**.

2. Drag **sales** and place it on the **Rows shelf**

Now , tableau will create a **bar chart** by **default**.

3. Now click on **show me** . then select the **treemap** to create rectangles for each **sub-category**.

For example : **the larger dark box = high sales**

— **The small light box = low sales**



mark card

Now the **rectangles are colored** by the

First class

Second class

Standard class

Same day

By the **controls size** .

5. Now click **undo** and remove **ship mode** . then
drag **profit** and place it on the **color**

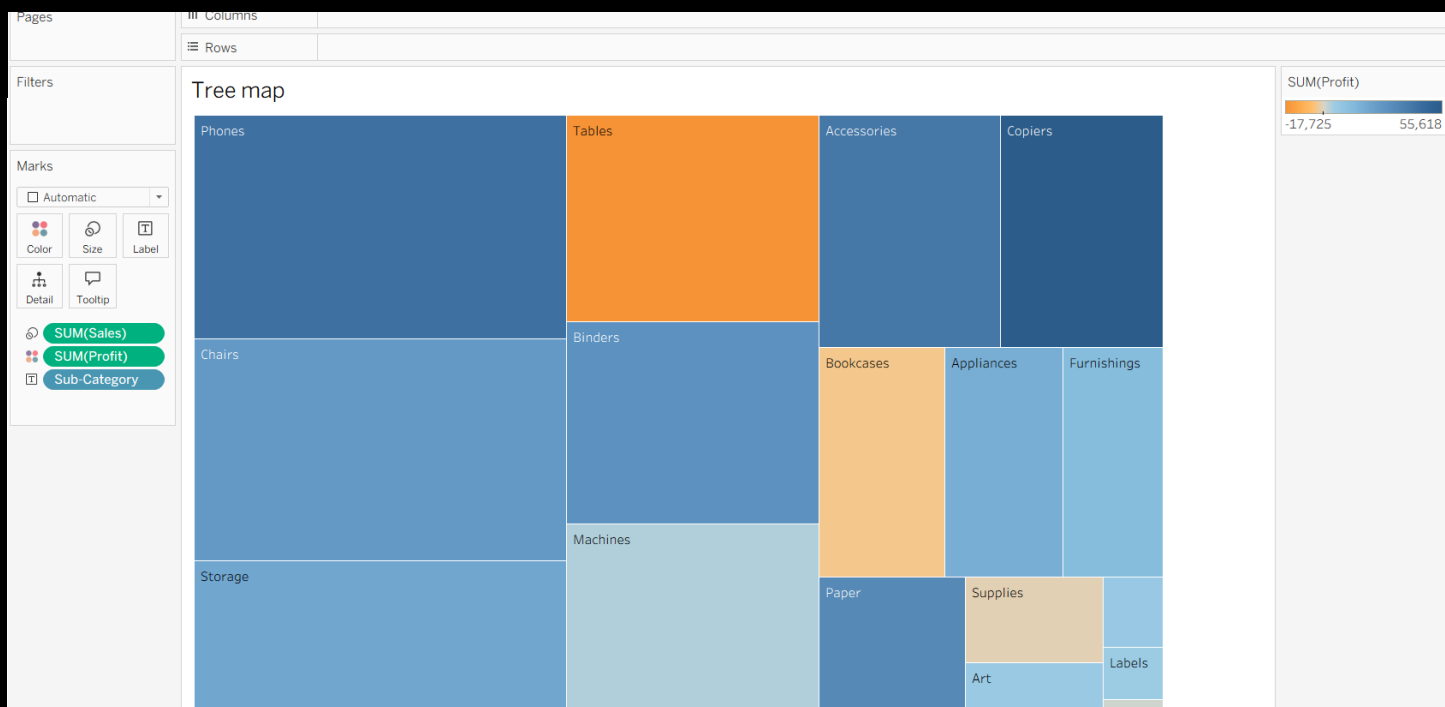
Size =sales

Color=profit

Output:

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—





ALPHA
— INNOVATORS —



1. Creating the **calculated field** as **zero**

* Now go to **analysis menu** and select **create calculated field**

*Name the field as **Zero** and add the formula **area**, as

Formula : 0

Click **okay**

2. Now **zero** will be added in the measures section

3. Drag **zero** to the **rows shelf**

4. click **size** in the **mark** and select the **pie** .
from the **toolbar** ,choose fit entire view to get the clear larger **pie**

right side of the first **Zero** on **rows shelf**.

6.click on the **SUM(zero)** in row ,then select **dual axis**.

Now both the pie charts overlap

7.click the second **SUM(zero)** ,in the **mark card**

Click **size** and reduce the size slightly and click **color** ,select the white color to get the clear **dount chart**

8.Now , click the first **SUM(zero)**on the **mark card** and Drag the **region to color**

Each region gets different color .

9. *Drag **region,sales** and place it on the **label** in **mark card**

10. * click the **second SUM(Zero)** in **mark card**.

*Drag **sales** to the **label**

*click **label**→**Text** and click the **three dots (...)**beside the **text editing**.

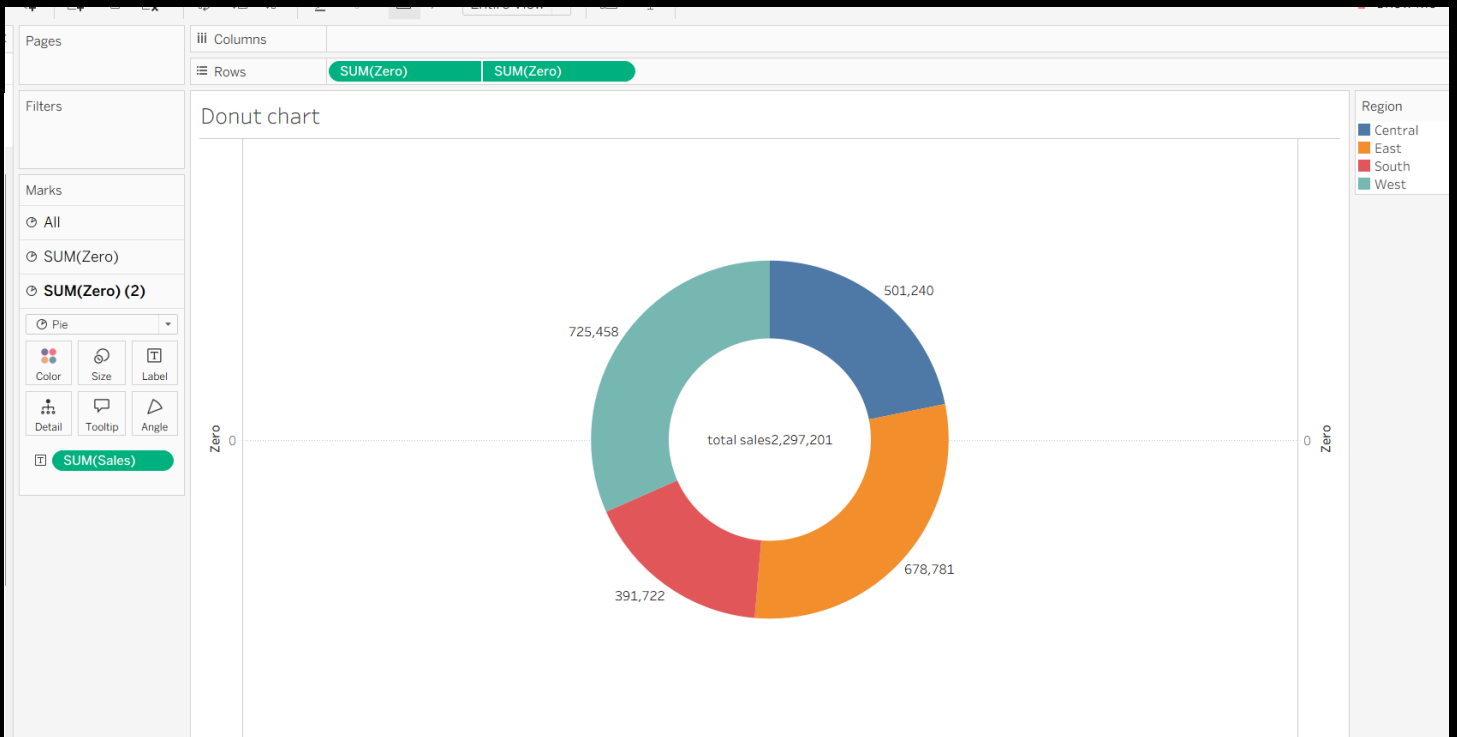
Type: **Total sales**

Select **ok**.

Output:

ALPHA
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ALPHA
— INNOVATORS —



1. Creating the basic map, from the data pane and Drag state to the workspace then the tableau automatically generates a map using latitude and longitude

2. Drag sales and place it on the color in mark card .

Now the map displays sales values with different color

3. Mark card ,click the automatic

Next select map or filled map

4. Map → Shows circles on locations and Filled Map → Fills states with colors.

and choose a color palette (purple)

Click ok

6. Drag state to label and again drag state to label . now each state shows its sales value

7. Drag region to the filters shelf .

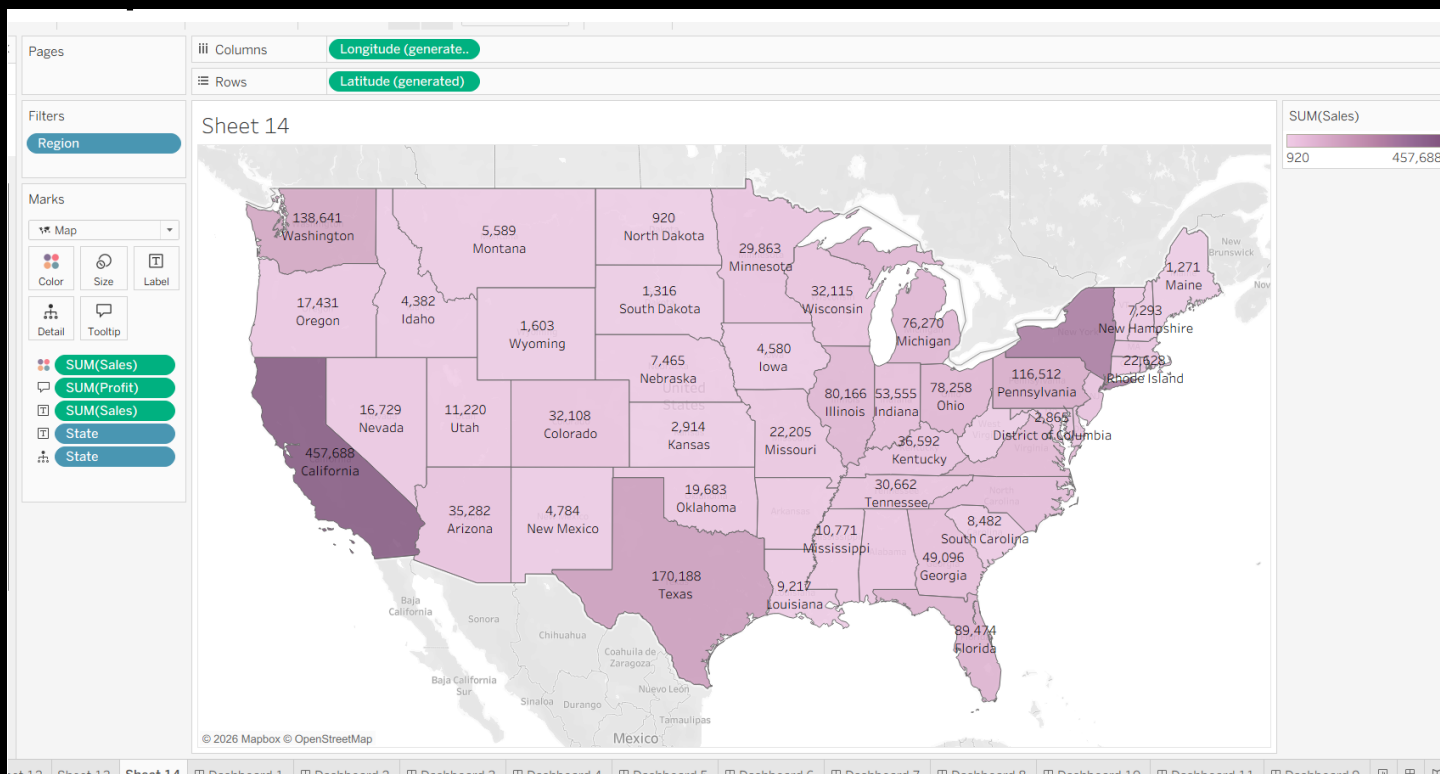
Select all region and click ok

8. Now click unknown right side down on the map and click it and then edit location to US(united state) and click ok

Then the map will appear to get output

ALPHA
— INNOVATORS —
—



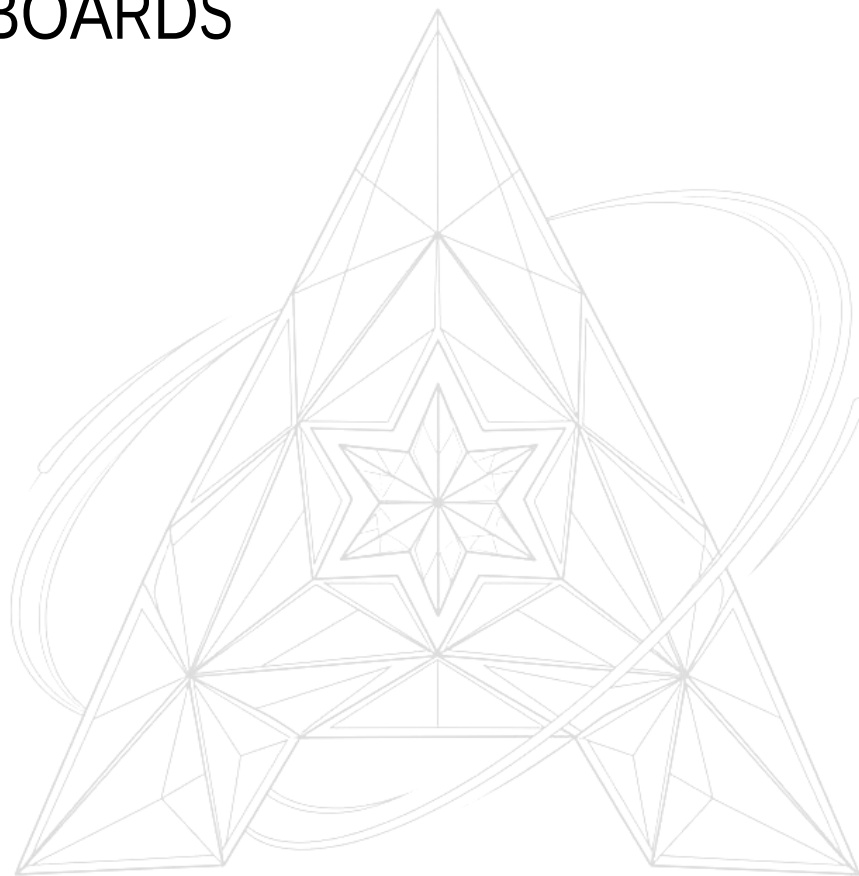


ALPHA

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DASHBOARDS



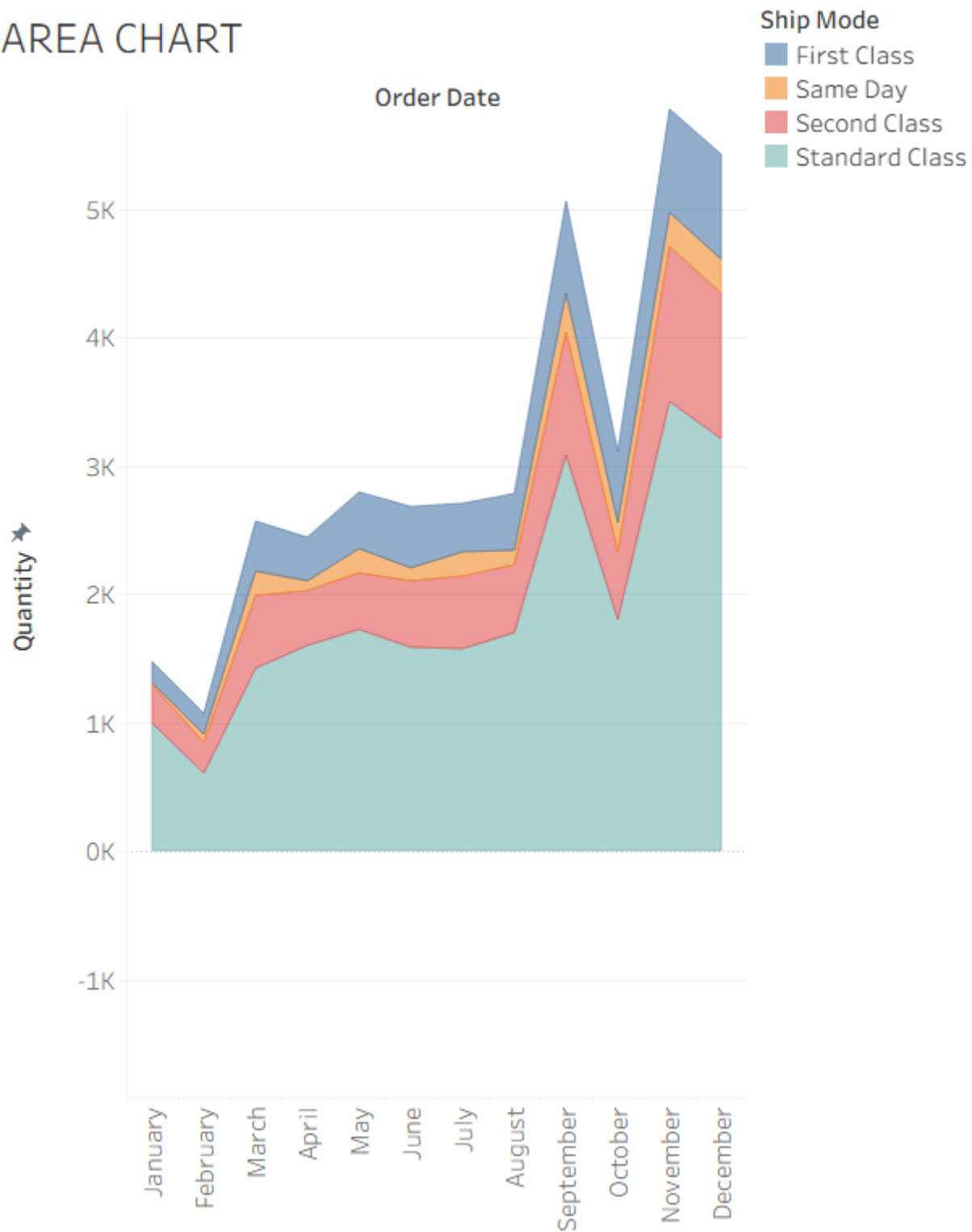
ALPHA

— INNOVATORS —



Dashboard 1

AREA CHART



BAR CHART

Region

West

Sales

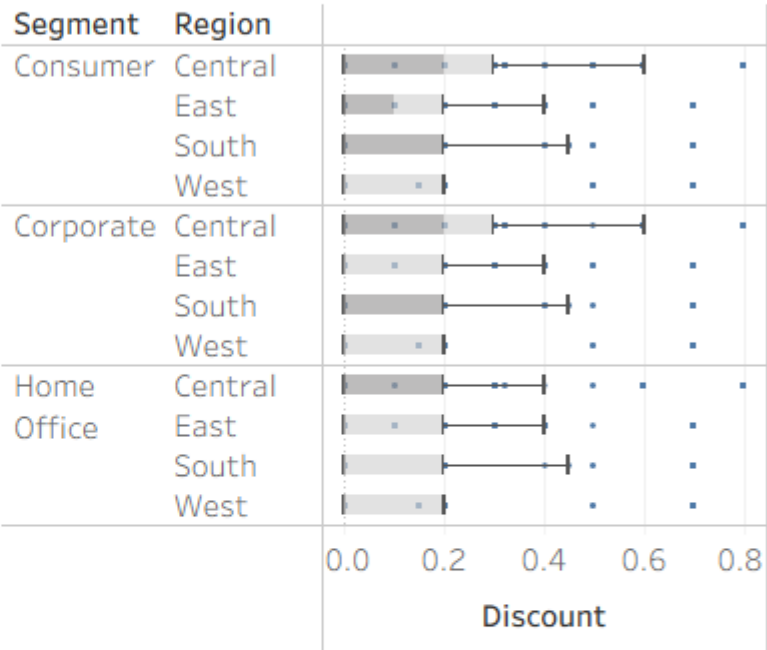
750K
700K
650K
600K
550K
500K
450K
400K
350K
300K
250K
200K
150K
100K
50K
0K



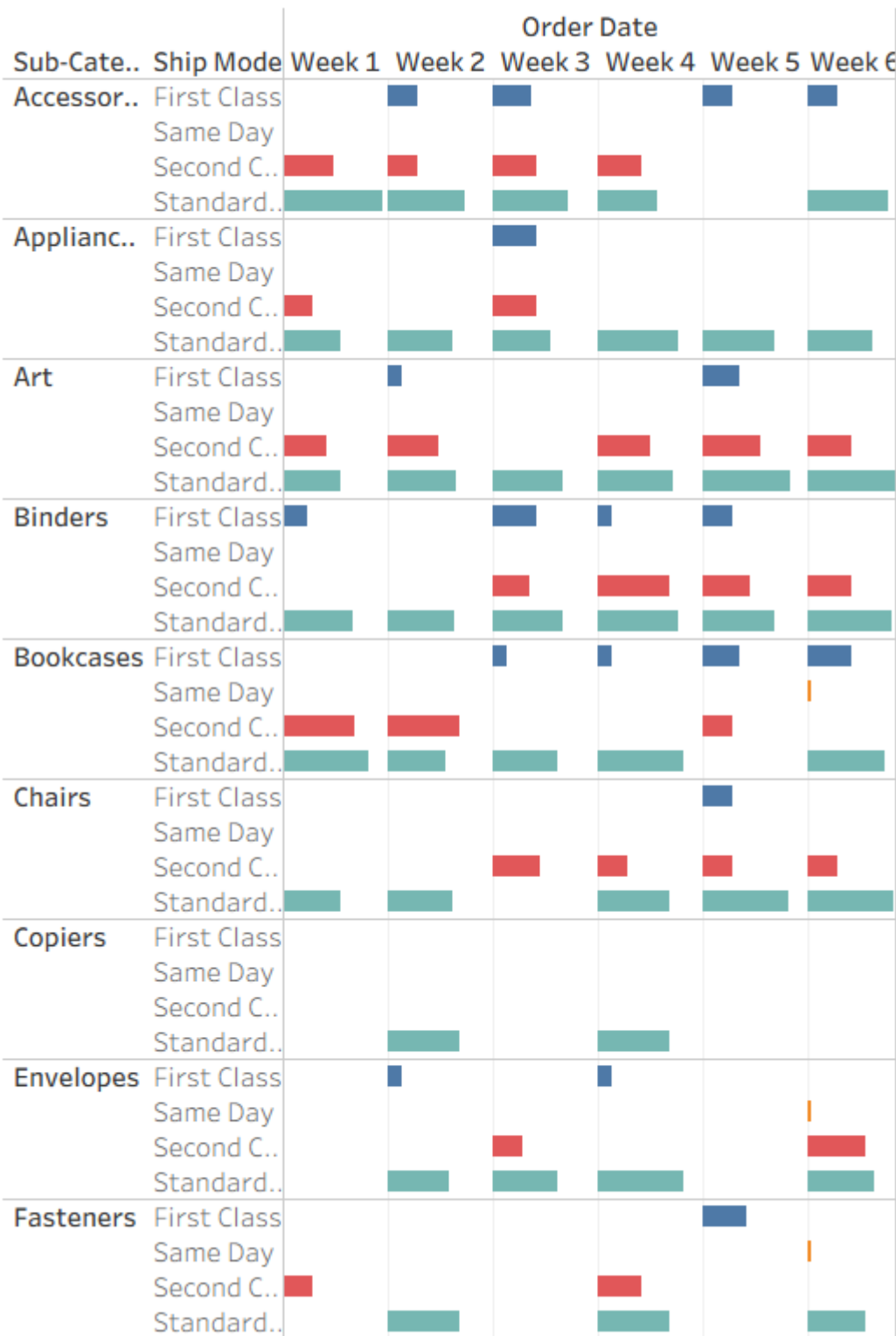
Ship Mode

- First Class
- Same Day
- Second Class
- Standard Class

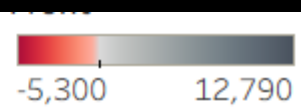
box plot



Gantt Chart



Heat Map

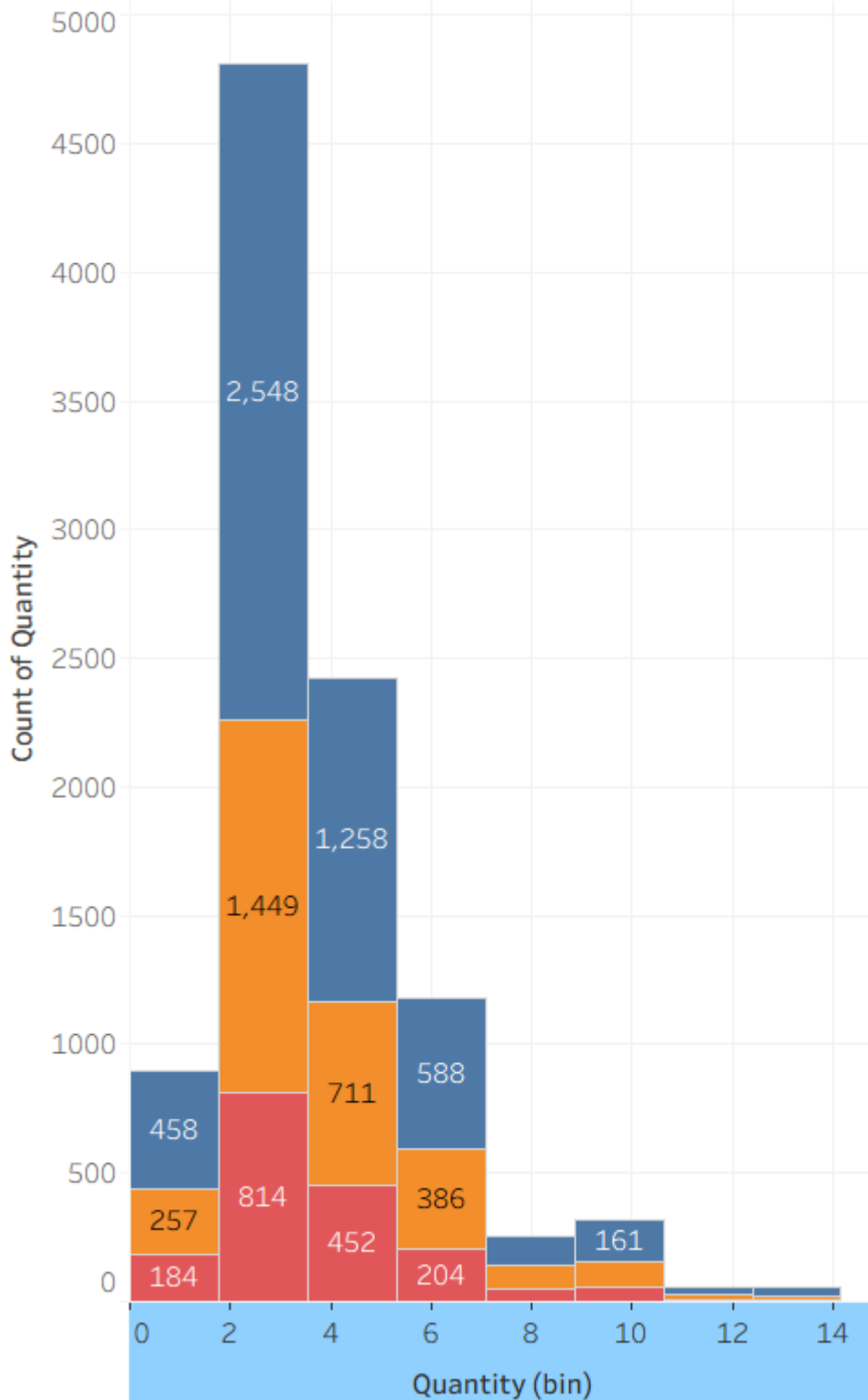


Region	Sub-Catego..	Segment		
		Consumer	Corporate	Home Office
Central	Accessories			
	Appliances			
	Art			
	Binders			
	Bookcases			
	Chairs			
	Copiers			
	Envelopes			
	Fasteners			
	Furnishings			
	Labels			
	Machines			
	Paper			
	Phones			
	Storage			
	Supplies			
	Tables			
East	Accessories			
	Appliances			
	Art			
	Binders			
	Bookcases			
	Chairs			
	Copiers			
	Envelopes			
	Fasteners			
	Furnishings			
	Labels			
	Machines			
	Paper			
	Phones			
	Storage			
	Supplies			
	Tables			
South	Accessories			
	Appliances			
	Art			

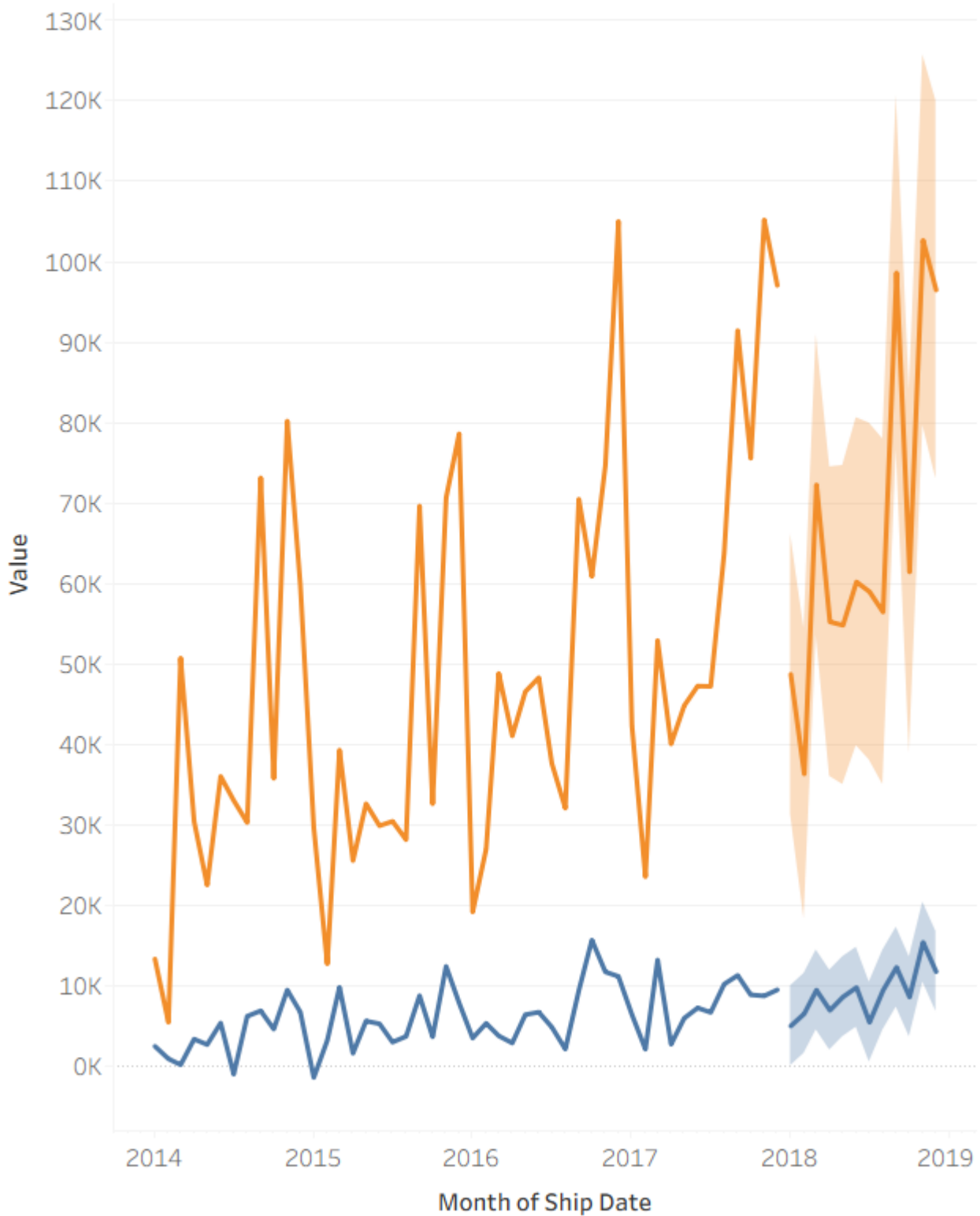
Histogram

Segment

- Consumer
- Corporate
- Home Office



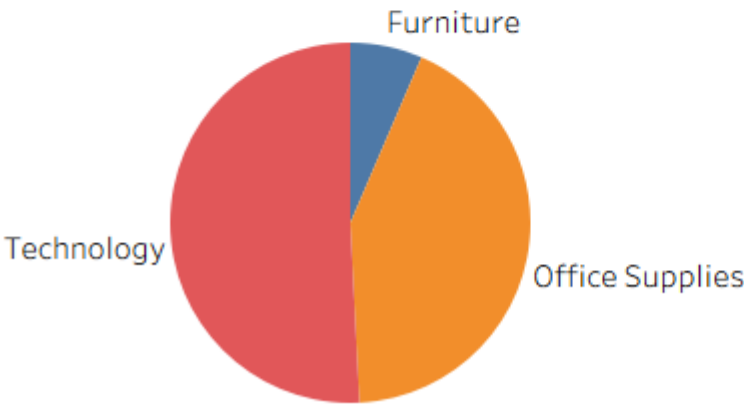
Line chart



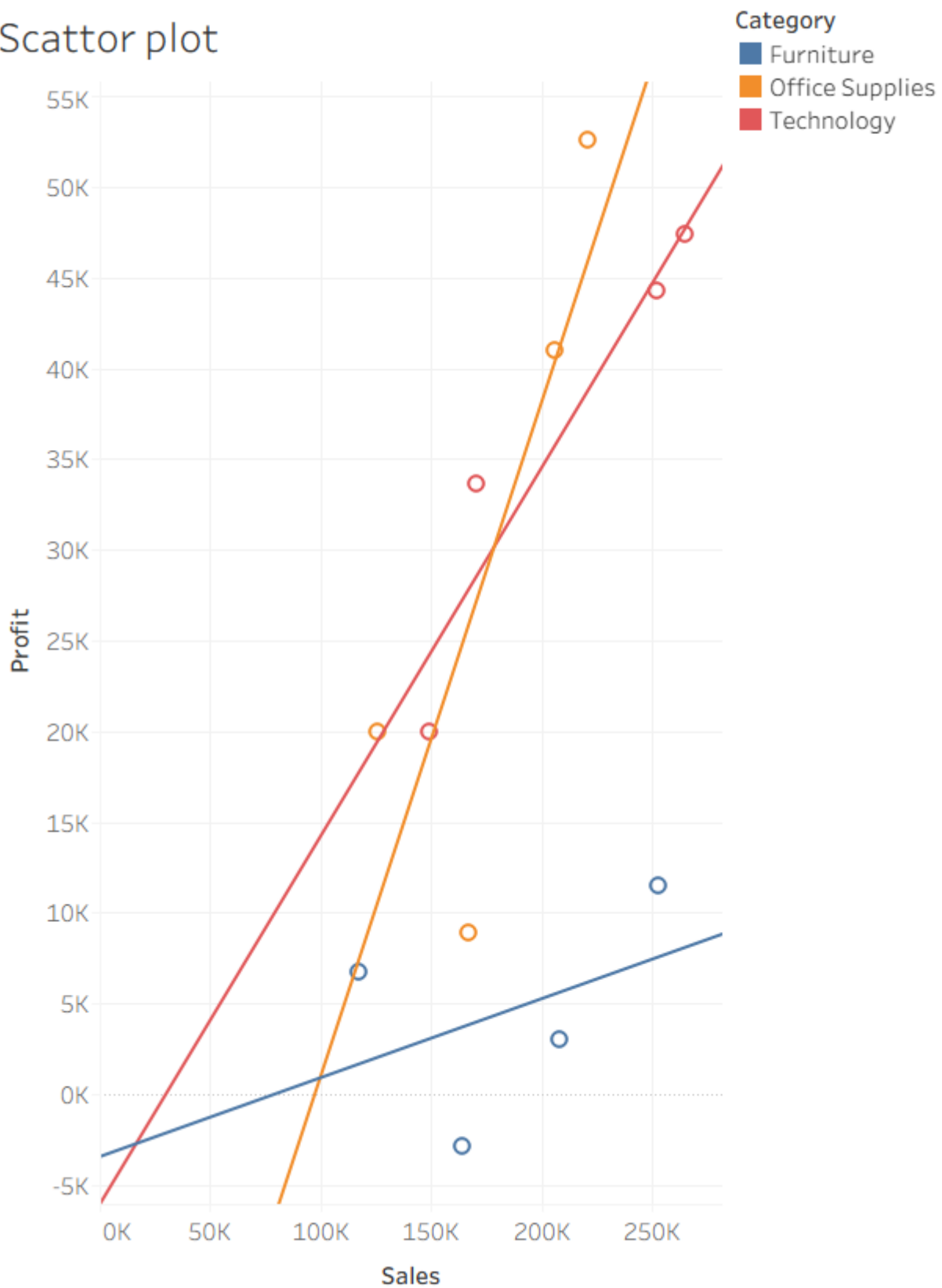
Sheet 8



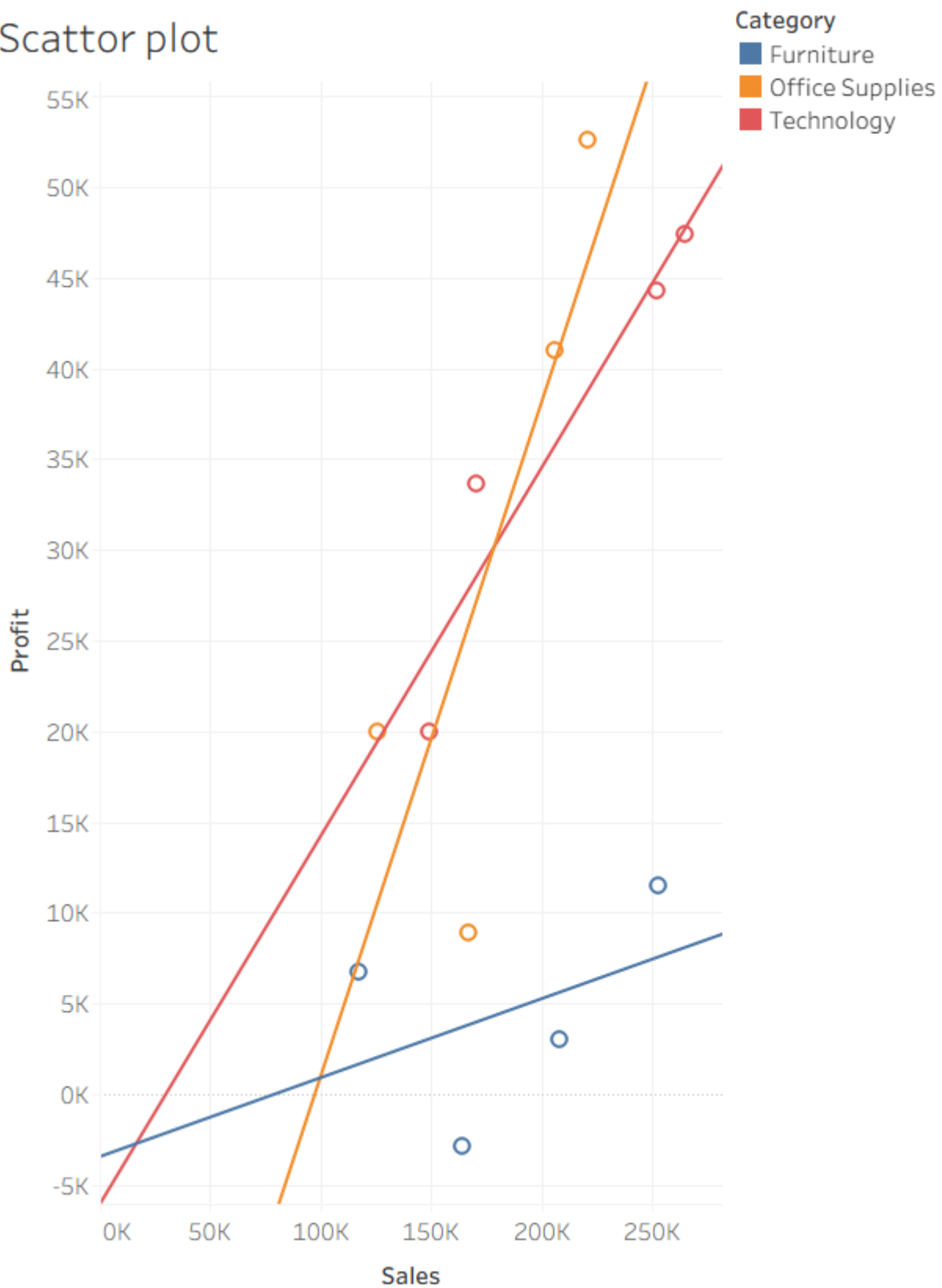
Pie Chart



Scatter plot

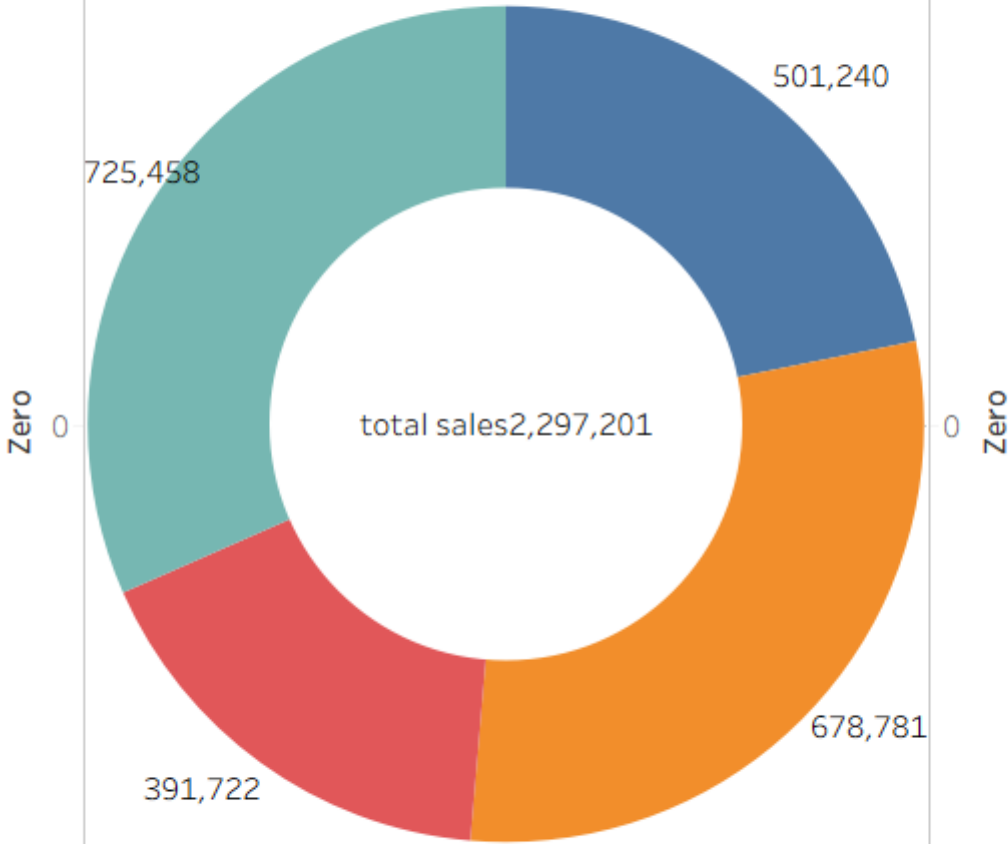


Scatter plot

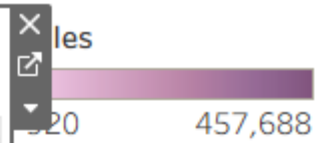
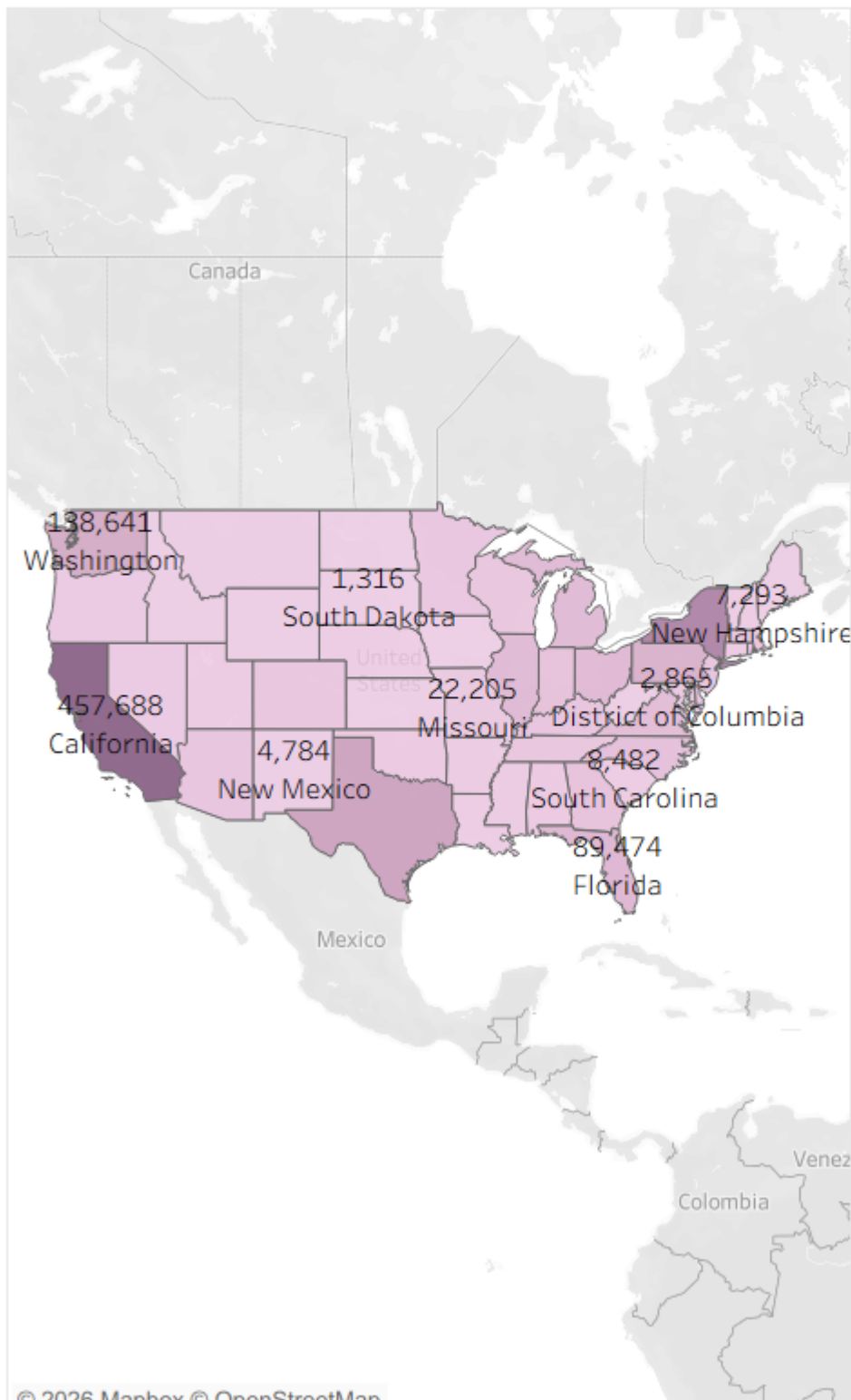


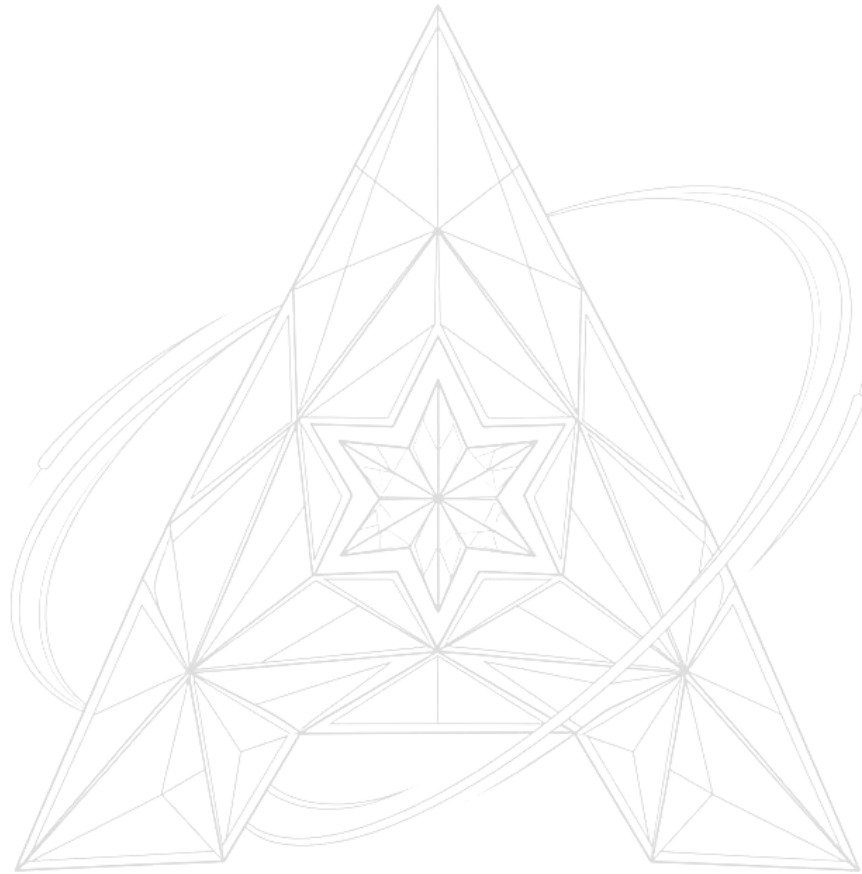
Donut chart

- Region
- Central
 - East
 - South
 - West



Sheet 14





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